

ECO211: Microeconomics I

Consumer Choice

Budget Constraint

Budget Constraint

- The **budget constraint** is the set of bundles that the consumer can afford, given the price of each good and her income.

- *Example:*

The budget set for good x (food) and y (clothing) is

$$p_x x + p_y y \leq I.$$

where p_x is the price of each unit of food;

p_y is the price of each unit of clothing;

I is the consumer's available income to spend on food and clothing.

Budget Constraint

- Bundles (x, y) that satisfy:
 - $p_x x + p_y y < I$
 - the consumer does not use all her income.
 - $p_x x + p_y y = I$
 - the consumer spends all her income.
- We refer to $p_x x + p_y y = I$ as the *budget line*.

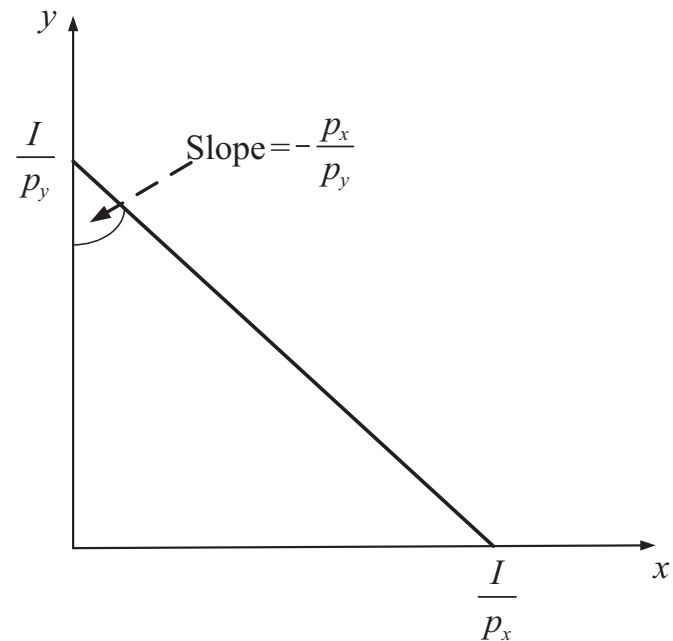
Budget Constraint

- Rearranging the budget line and solving for y ,

$$p_y y = I - p_x x,$$
$$y = \underbrace{\frac{I}{p_y}}_{\text{Vertical intercept}} - \underbrace{\frac{p_x}{p_y}}_{\text{Slope}} x.$$

- Setting $y = 0$, and solving for x we find the horizontal intercept at

$$p_x x + p_y 0 = I,$$
$$x = \frac{I}{p_x}.$$



Budget Constraint

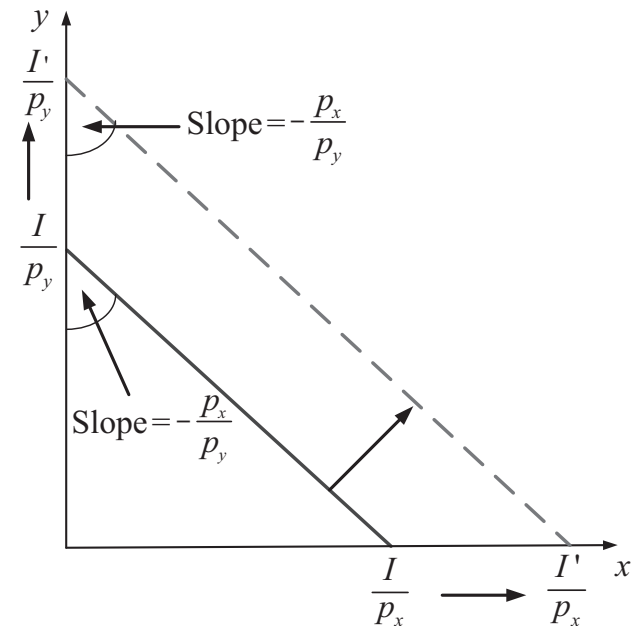
- The slope of the budget line, tells us how many units of y the consumer must give up to buy 1 more unit of x
- If $p_x = 10$ and $p_y = 20$, the slope is

$$-\frac{p_x}{p_y} = -\frac{10}{20} = -\frac{1}{2}.$$

- The consumer must give up $1/2$ units of good y to acquire 1 more unit of good x , because good y is twice as expensive as good x .
- Alternatively, she must give up 1 unit of good y to purchase 2 more units of good x .

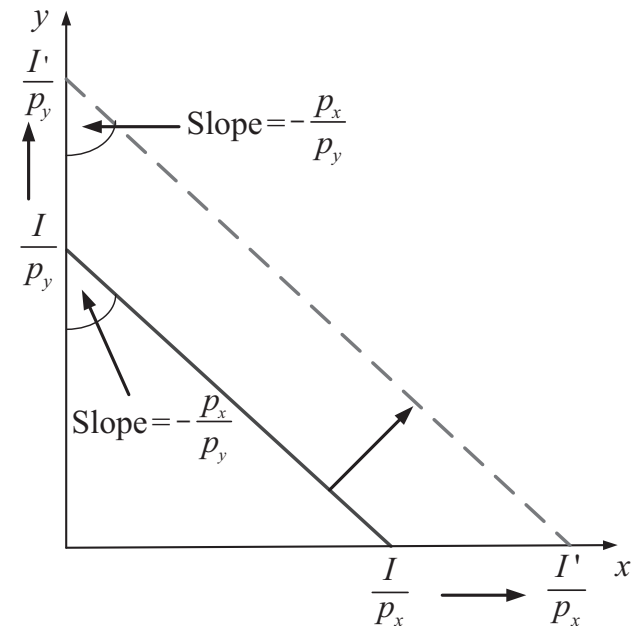
Budget Constraint

- *Changes in income:*
 - An *increase* in income from I to I' , where $I' > I$, shifts the budget line outward in a parallel fashion.
 - As income increase, she can afford a larger set of bundles.



Budget Constraint

- *Changes in income* (cont.):
 - A *decrease* in income produces the opposite, a shifting inward in a parallel fashion.

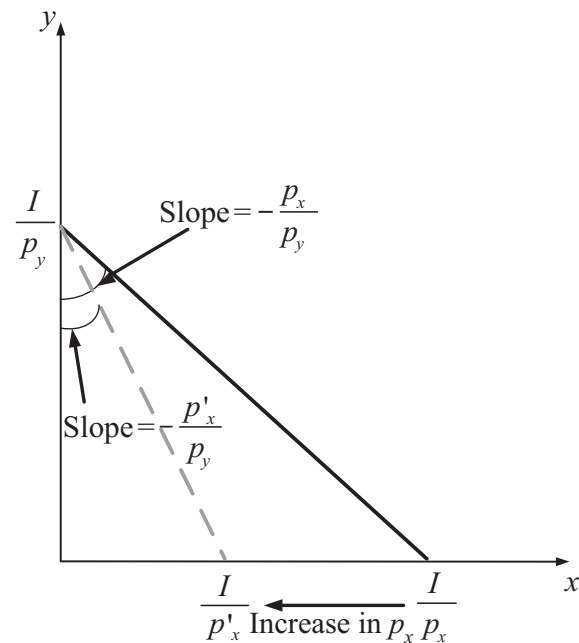


Budget Constraint

- *Changes in prices:*

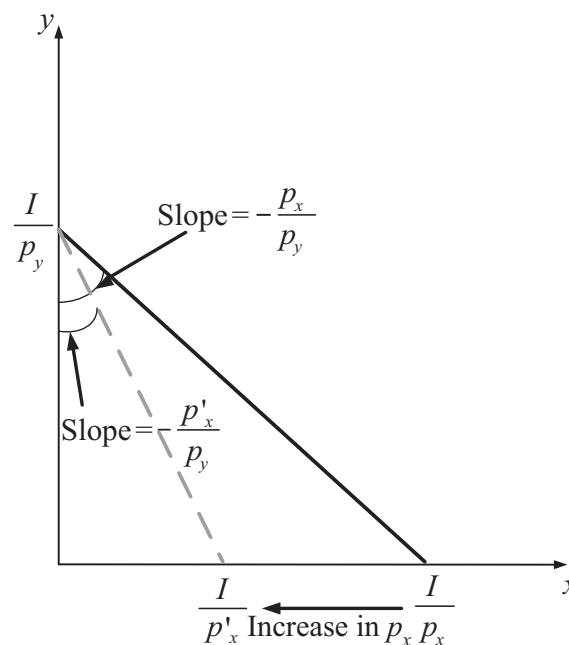
- An *increase* in the price of one good, such as p_x , pivots the budget line inward.

- The vertical intercept $\frac{I}{p_y}$ is unaffected.
- The horizontal intercept $\frac{I}{p_x}$ moves leftward.



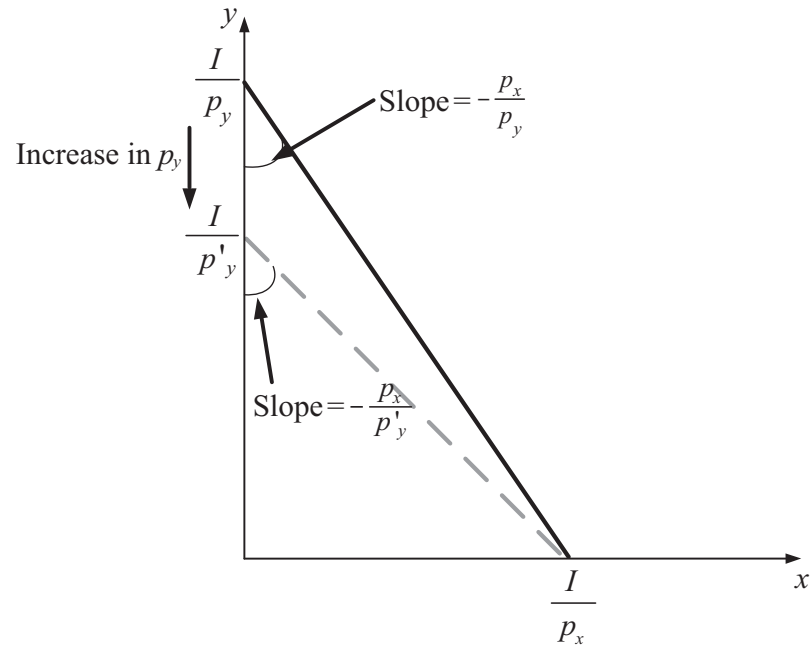
Budget Constraint

- *Changes in prices* (cont.):
 - An *increase* in the price of one good, such as p_x , pivots the budget line inward.
 - The consumer faces a more expensive good, shrinking the set of bundles she can afford.
 - A *decrease* of p_x has the opposite effect, moving the horizontal intercept rightward.



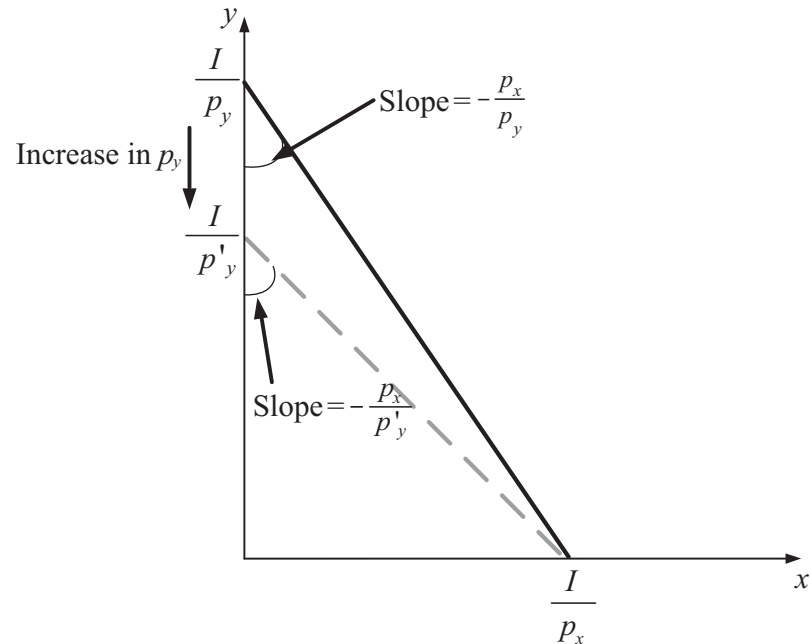
Budget Constraint

- *Changes in prices* (cont.):
 - A similar argument applies if the price of good y , p_y , increases.
 - The horizontal intercept $\frac{I}{p_x}$ is unaffected.
 - The vertical intercept $\frac{I}{p_y}$ moves down.



Budget Constraint

- *Changes in prices* (cont.):
 - A similar argument applies if the price of good y , p_y , increases.
 - A decrease in p_y moves the vertical intercept up.



Budget Constraint

- *Query:*

What would happen if both income and the price of all goods were doubled?

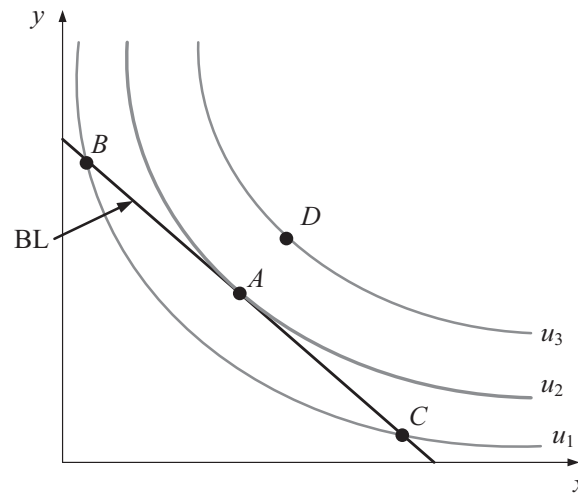
- The budget line is unaffected!
 - The vertical intercept of the budget line would become $\frac{2I}{2p_y}$, which simplifies to $\frac{I}{p_y} \rightarrow$ no change in its position.
 - The horizontal intercept is now $\frac{2I}{2p_x}$, reducing to $\frac{I}{p_x}$.
 - And the slope does not change either, $-\frac{2p_x}{2p_y} = -\frac{p_x}{p_y}$.

This argument applies to any common increase (decrease) in all prices and income.

Utility Maximization Problem

Utility Maximization Problem

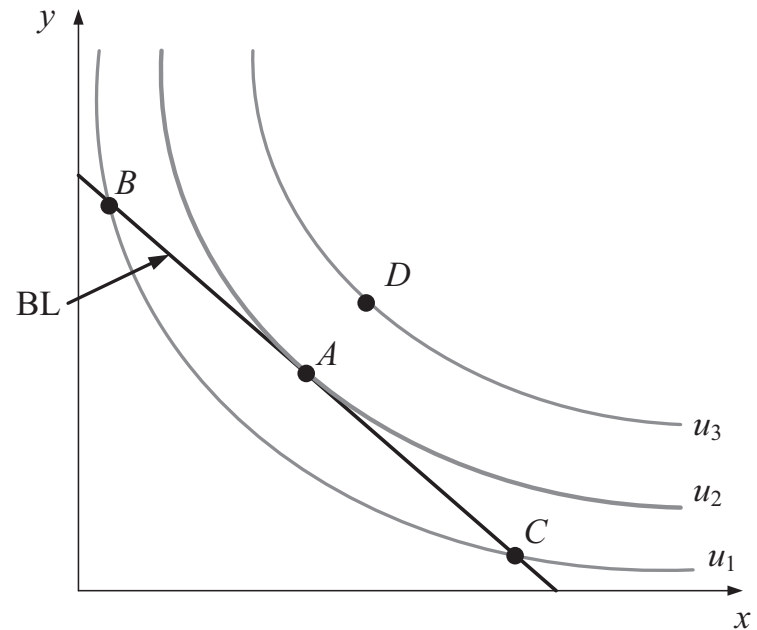
- The process by which the consumer chooses utility-maximizing bundles, that are bundles that maximize her utility among all of those she can afford.



- Let's test if points $A - D$ are utility-maximizing for the consumer.

Utility Maximization Problem

- Bundles C and B cannot be optimal. She reaches u_1 spending all her income, $p_x s + p_y = I$. But at bundle A (with same spending) she reaches a higher utility u_2 , $u_2 > u_1$.
- Bundle D cannot be optimal. It yields a higher utility than A , but it is unaffordable.
- Only bundle A is optimal, where the budget line and indifference curves are *tangent* each other.



Utility Maximization Problem

- This tangency condition requires that the slope of the budget line at bundle A , $\frac{p_x}{p_y}$, is equal to the slope of the indifference curve,

$$MRS = \frac{MU_x}{MU_y}.$$

- Therefore, utility-maximizing bundles must satisfy

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y} \text{ or after rearranging } \frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

- This condition states that marginal utility per dollar spent on the last unit of good x must be equal to that of good $y \rightarrow$ *bang for the buck* must coincide across all goods.
- If $\frac{MU_x}{p_x} > \frac{MU_y}{p_y}$, the consumer would obtain a larger bang for the buck from x than y , providing incentives to spend more money on x .

Lagrange Method to Solve UMP

- We use the Lagrange method to show that the tangency condition must be satisfied at the optimum of the UMP. The UMP can be expressed as

$$\begin{aligned} & \max_{x,y} u(x, y) \\ & \text{subject to } p_x x + p_y y = I. \end{aligned}$$

- We use the budget line $p_x x + p_y y = I$, rather than the budget constraint $p_x x + p_y y \leq I$, because the consumer will always spend all her available income.
- The consumer faces a “constrained maximization problem.”

A. Lagrange Method to Solve UMP

- Constrained maximization problems are often solved by setting up a Lagrangian function,

$$\mathcal{L}(x, y; \lambda) = u(x, y) + \lambda[I - p_x x - p_y y],$$

where λ represents the Lagrange multiplier, which multiplies the budget line.

- To solve this problem, we take FOP with respect to x , y , and λ ,

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0, \text{ and}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - p_x x - p_y y = 0.$$

A. Lagrange Method to Solve UMP

- The first and the second conditions can be rearranged to

$$\frac{MU_x}{p_x} = \lambda \text{ and } \frac{MU_y}{p_y} = \lambda.$$

- Because both conditions are equal to λ , we obtain

$$\frac{MU_x}{p_x} = \lambda = \frac{MU_y}{p_y}$$

This is the “bang for the buck” coinciding across goods.

- Alternatively, this condition can be expressed as

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y},$$

which coincides with the tangency condition used in the previous analysis.

Utility Maximization Problem

- *Procedure to solve the UMP:*

1. Set the tangency condition as $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$. Cross-multiply and simplify.
2. If the expression for the tangency condition:
 - a. Contains both unknowns (x and y), solve for x , and insert the resulting expression into the budget line $p_x x + p_y y = I$.
 - b. Contains only one unknown (x or y), solve for that unknown, and insert the result into the budget line $p_x x + p_y y = I$.

Utility Maximization Problem

- *Procedure to solve the UMP* (cont.):

2. If the expression for the tangency condition:

- c. Contains no good x or y , compare $\frac{MU_x}{p_x}$ against $\frac{MU_y}{p_y}$.
- If $\frac{MU_x}{p_x} > \frac{MU_y}{p_y}$, set good $y = 0$ in the budget line and solve for good x (corner solution where the consumer purchases only good x).
 - If $\frac{MU_x}{p_x} < \frac{MU_y}{p_y}$, set $x = 0$ in the budget line and solve for y (corner solution where she purchases only good y).

Utility Maximization Problem

- *Procedure to solve the UMP* (cont.):
 3. If, in step 2, you find that one of the goods is consumed in negative amounts (e.g., $x = -2$), then set the amount of this good equal to 0 on the budget line (e.g., $p_x 0 + p_y y = I$), and solve for the remaining good.
 4. If you haven't found the values for all the unknowns, use the tangency conditions from step 1 to find the remaining unknown.

Utility Maximization Problem

- *Example 1: UMP with interior solutions—I.*

- Consider an individual with Cobb-Douglas utility function

$$u(x, y) = xy.$$

facing $p_x = 20$, $p_y = 40$, and $I = 800$.

- *Step 1.* We use the tangency condition to find optimal bundle

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y},$$
$$\frac{y}{x} = \frac{20}{40} \Rightarrow \frac{y}{x} = \frac{1}{2},$$

$$2y = x.$$

This result contains both x and y , so we move to step 2a.

Utility Maximization Problem

- *Example 1* (continued):

- *Step 2a.* From the budget line, $20x + 40y = 800$.

Inserting $2y = x$ into the budget line,

$$20(\underbrace{2y}_x) + 40y = 800,$$

$$80y = 800,$$

$$y = \frac{800}{80} = 10 \text{ units.}$$

Because the consumer purchases 10 units of y , we move to step 4 (recall that we only need to stop at step 3 if x or y are negative in step 2).

- *Step 4.* To find the optimal consumption of x , we use the tangency condition $x = 2y = 2 \times 10 = 20$ units.

Utility Maximization Problem

- *Example 1* (continued):
 - *Summary.* The optimal consumption bundle is (20,10).
The slope of the indifference curve, $\frac{y}{x} = \frac{10}{20} = \frac{1}{2}$, coincides with that of the budget line, $\frac{p_x}{p_y} = \frac{1}{2}$.

Utility Maximization Problem

- *Example 2: UMP with interior solutions–II.*
 - Consider an individual with Cobb-Douglas utility function

$$u(x, y) = x^{1/3} y^{2/3}$$

facing $p_x = 10$, $p_y = 20$, and $I = 100$.

Before using the tangency condition $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$, we first find

$$\frac{MU_x}{MU_y} = \frac{\frac{1}{3} x^{\frac{1}{3}-1} y^{\frac{2}{3}}}{\frac{2}{3} x^{\frac{1}{3}} y^{\frac{2}{3}-1}} = \frac{\frac{1}{3} x^{-\frac{2}{3}} y^{\frac{2}{3}}}{\frac{2}{3} x^{\frac{1}{3}} y^{-\frac{1}{3}}} = \frac{y^{\frac{2}{3}+\frac{1}{3}}}{2x^{\frac{1}{3}+\frac{2}{3}}} = \frac{y}{2x}.$$

Utility Maximization Problem

- *Example 2* (continued):

- *Step 1.* We use the tangency condition $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$

$$\frac{y}{2x} = \frac{10}{20},$$

$$y = x.$$

This result contains x and y , so we move to step 2a.

Utility Maximization Problem

- *Example 2* (continued):
 - *Step 2a.* Inserting $y = x$ into the budget line,

$$10x + 20y = 100,$$

$$20(y) + 20y = 100,$$

$$30y = 100,$$

$$y = \frac{100}{30} = 3.33 \text{ units.}$$

Utility Maximization Problem

- *Example 2* (continued):
 - *Step 4.* The optimal consumption of x can be found by using the tangency condition
$$y = x = 3.33 \text{ units.}$$
 - *Summary.* The optimal consumption bundle is (3.33, 3.33).

Utility Maximization Problem

- *Example 2* (continued):

- We can find the budget shares of each good, that is the % of income the consumer spends on good x and good y :

$$\frac{p_x x}{I} = \frac{10 \times 3.33}{100} = \frac{1}{3},$$
$$\frac{p_y y}{I} = \frac{20 \times 3.33}{100} = \frac{2}{3}.$$

which coincides with the exponent of each good in the Cobb-Douglas utility function $u(x, y) = x^{1/3} y^{2/3}$.

- This result can be generalized to all types of Cobb-Douglas utility functions $u(x, y) = Ax^\alpha y^\beta$, where $A, \alpha, \beta > 0$.
 - The budget share of good x is α , and of good y is β .

Utility Maximization Problem

- *Example 3: UMP with corner solutions.*
 - Consider a consumer with utility function $u(x, y) = xy + 7x$, and facing $p_x = 1$, $p_y = 2$, and $I = 10$.
 - *Step 1.* Using the tangency condition $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$,

$$\frac{y + 7}{x} = \frac{1}{2},$$

$$2y + 14 = x.$$

This result contains x and y , so we move to step 2a.

Utility Maximization Problem

- *Example 3* (continued):
 - *Step 2.* Inserting $2y + 14 = x$ into the budget line

$$x + 2y = 10,$$

$$\underbrace{(2y + 14) + 2y}_{x} = 10,$$

$$4y = -4,$$

$$y = -1.$$

Utility Maximization Problem

- *Example 3* (continued):

- *Step 3.* Because the amounts of x and y cannot be negative, the consumer would like to reduce her consumption of good y as much as possible (i.e., $y = 0$). Inserting this result into the budget line

$$x + (2 \times 0) = 10 \rightarrow x = 10 \text{ units.}$$

- *Summary.* We have found a corner solution, where the consumer uses all her income to purchase good x alone.
- Graphically, her optimal budget $(x, y) = (10, 0)$ is located at the horizontal intercept of her budget line.

Utility Maximization Problem

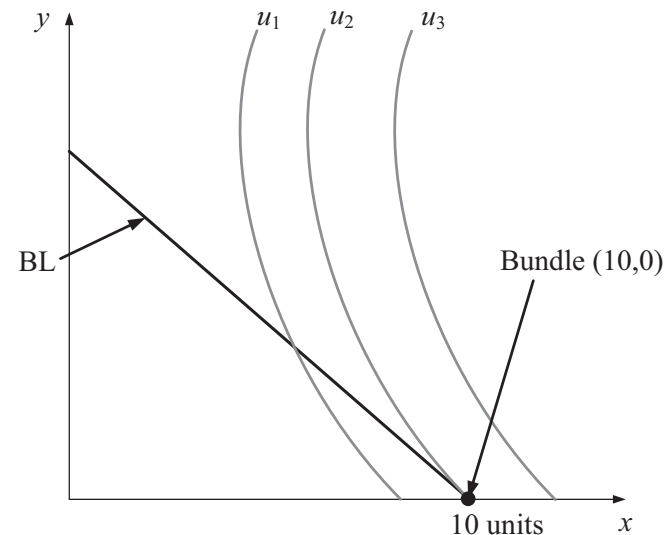
- *Example 3* (continued):

- At the corner solution, the tangency condition does not hold,

$$\frac{MU_x}{p_x} \neq \frac{MU_y}{p_y} \Rightarrow \frac{y+7}{1} \neq \frac{x}{2}.$$

At $(x, y) = (10, 0)$, $\frac{0+7}{1} > \frac{10}{2}$.

- This should induce the consumer to increase her consumption of x and decrease that of y .
- But, since she is at $y = 0$, she cannot longer decrease her consumption of y .



UMP in Extreme Scenarios

UMP in Extreme Scenarios

- Goods are regarded as perfect substitutes:
 - Consider two brands of mineral water. This utility function takes the form $u(x, y) = ax + by$, where $a, b > 0$.
 - In this scenario, $\frac{MU_x}{MU_y} = \frac{a}{b}$.
 - Three cases can emerge:
 1. $\frac{a}{b} > \frac{p_x}{p_y}$.
 2. $\frac{a}{b} < \frac{p_x}{p_y}$.
 3. $\frac{a}{b} = \frac{p_x}{p_y}$.

UMP in Extreme Scenarios

- Goods are regarded as perfect substitutes (cont.):
 1. If $\frac{a}{b} > \frac{p_x}{p_y}$, the IC is steeper than the budget line, producing a corner solution. The consumer spends all income on x .

Using the “bang for the buck” approach:

$$\frac{a}{p_x} > \frac{b}{p_y},$$

the bang for the buck from x is larger than that of y . So she consumer would like to increase her consumption of x while decreasing that of y .

UMP in Extreme Scenarios

- Goods are regarded as perfect substitutes (cont.):
 2. If $\frac{a}{b} < \frac{p_x}{p_y}$, a corner solution exists, where the consumer spends all her income on good y .

The optimal consumption bundle lies on the vertical intercept of the budget line.

UMP in Extreme Scenarios

- Goods are regarded as perfect substitutes (cont.):
 3. If $\frac{a}{b} = \frac{p_x}{p_y}$, the slope of the indifference curves and the budget line coincide, yielding a complete overlap.

Tangency occurs at all points of the budget line → a continuum of solutions exists, any bundle (x, y) satisfying $p_x x + p_y y = I$ is utility maximizing.

UMP in Extreme Scenarios

- Goods are regarded as perfect complements:
 - Consider cars and petrol. This utility function takes the form $u(x, y) = A \min\{ax, by\}$, where $A, a, b > 0$.
 - The ICs are L-shaped, and have a kink at a ray from the origin with slope a/b .
 - The MRS of this function is undefined.
 - We cannot use the tangency condition.
 - Optimal bundles require to identify bundles for which we cannot increase the consumer's utility given her budget constraint.

UMP in Extreme Scenarios

- Goods are regarded as perfect complements (cont.):
 - She consumes the bundle at the kink of her IC where it intersects her budget line.
 - Mathematically, it requires
 - $ax = by \Rightarrow y = \frac{b}{a}x$, for the bundle to be at the kink;
 - $p_x x + p_y y = I$, for the bundle to be on the budget line.
 - We have system of two equations and two unknowns.
 - Inserting the first equation into the second,

$$p_x x + p_y \underbrace{\frac{a}{b}x}_y = I \Rightarrow x = \frac{I}{p_x + p_y \frac{a}{b}} = \frac{bI}{bp_x + ap_y}.$$

UMP in Extreme Scenarios

- Goods are regarded as perfect complements (cont.):

- The optimal amount of y becomes

$$y = \frac{a}{b} + \underbrace{\frac{bI}{bp_x + ap_y}}_x = \frac{aI}{bp_x + ap_y}.$$

- If $a = b = 2$ (when the individual needs to consume the same amount of each good), and $p_x = 10$, $p_y = 20$, and $I = 100$, the optimal consumptions of x and y are

$$x = \frac{bI}{bp_x + ap_y} = \frac{2 \times 100}{(2 \times 10) + (2 \times 20)} = \frac{10}{3} \text{ units,}$$

$$y = \frac{aI}{bp_x + ap_y} = \frac{2 \times 100}{(2 \times 10) + (2 \times 20)} = \frac{10}{3} \text{ units.}$$

Revealed Preferences

Revealed Preference

- Previously, we have analyzed how to find optimal bundles, assuming we observe consumer's preferences represented with her utility function.
- *What if we only know which choices she made when facing different combinations of prices and income?*
- We still can check if the consumer made optimal choices using the **Weak Axiom of Revealed Preferences (WARP)**.

Revealed Preference

- Consider:
 - $A = (x_A, y_A)$ be the optimal bundle when facing initial prices and income (p_x, p_y, I) .
 - $B = (x_B, y_B)$ be the optimal bundle when facing final prices and income (p'_x, p'_y, I') .

Revealed Preference

- **Weak Axiom of Revealed Preference (WARP).** If optimal consumption bundles A and B are both affordable under initial prices and income (p_x, p_y, I) , then bundle A cannot be affordable under final prices and income (p'_x, p'_y, I') :
 - If $p_x x_A + p_y y_A \leq I$ and $p_x x_B + p_y y_B \leq I$,
 - then $p'_x x_A + p'_y y_A > I'$.

Revealed Preference

- Weak Axiom of Revealed Preference (WARP) (cont.).
 - If both bundles are initially affordable, and the consumer selects A , she is “revealing” her preference for A over B .
 - WARP requires A is not affordable under final prices and income, otherwise the consumer should still select the original bundle A .
- Think of WARP as a *consistency* requirement in consumer’s choices when facing different prices and incomes.

Revealed Preference

- *Checking for WARP:*

1. *Checking the premise.* Check if bundles A and B are initially affordable $\rightarrow$ they lie on or below the budget line, BL , (p_x, p_y, I) .
 - 1a. If step 1 holds, move to step 2.
 - 1b. If step 1 does not hold, stop. We can only claim that the consumer choices *do not violate* WARP.
2. *Checking the conclusion.* Check that bundle A is no longer affordable $\rightarrow$ it lies strictly above the final budget line BL' , (p'_x, p'_y, I') .
 - 2a. If step 2 holds, WARP is *satisfied*.
 - 2b. If step 2 does not hold, WARP is *violated*.

Revealed Preference

- *Example: Testing for WARP.*
 - Consider a change in the budget line, from BL to BL' , due to a simultaneous decrease in p_x and I .
 - Initially, A is chosen and later B is chosen.

Revealed Preference

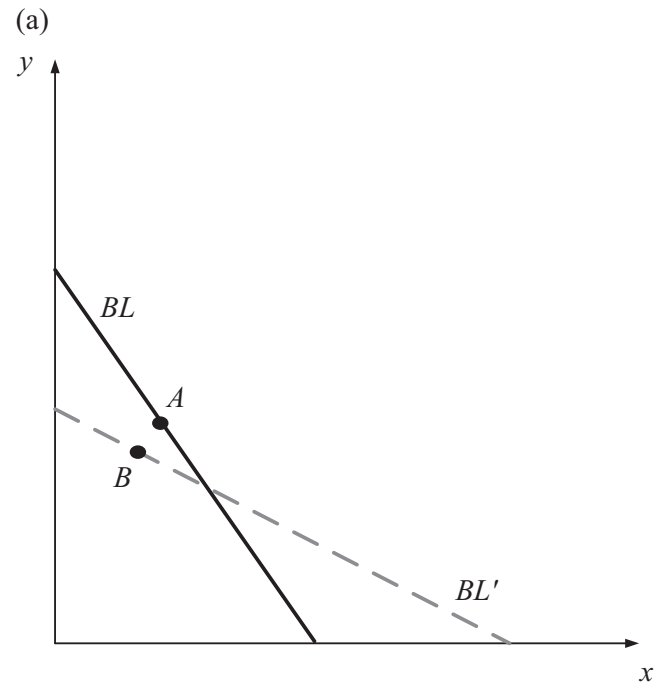
- *Example* (continued):
 - Scenario (a). WARP *is not violated*.

Step 1 holds. Bundles A and B are affordable under BL :

- A lies on BL .
- B lies strictly below BL .

Step 2 holds. Bundle A is unaffordable under BL' :

- A lies strictly above BL' .



Revealed Preference

- *Example* (continued):
 - Scenario (b). *WARP is violated.*

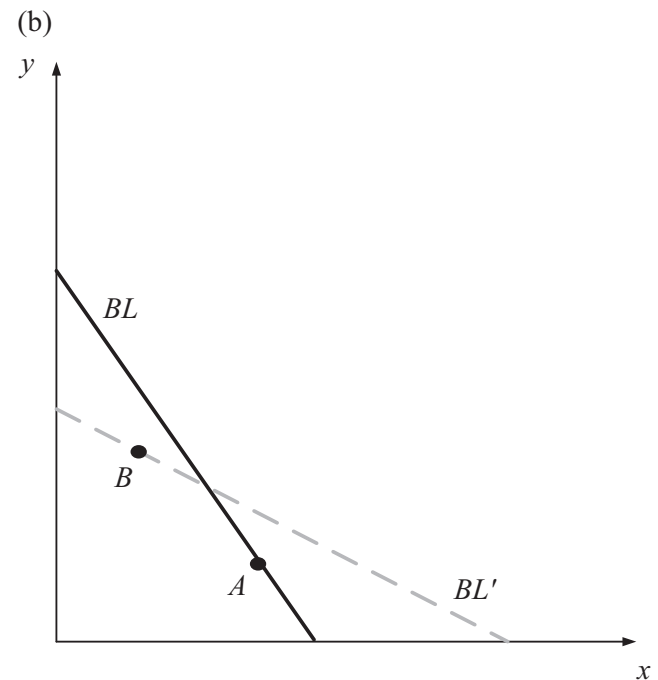
Step 1 holds. Bundles A and B are affordable under BL :

- A lies on BL .
- B lies strictly below BL .

Step 2 does not hold. Bundle A is affordable under BL' :

- A lies strictly below BL' .

The consumer is not consistent in her choices.



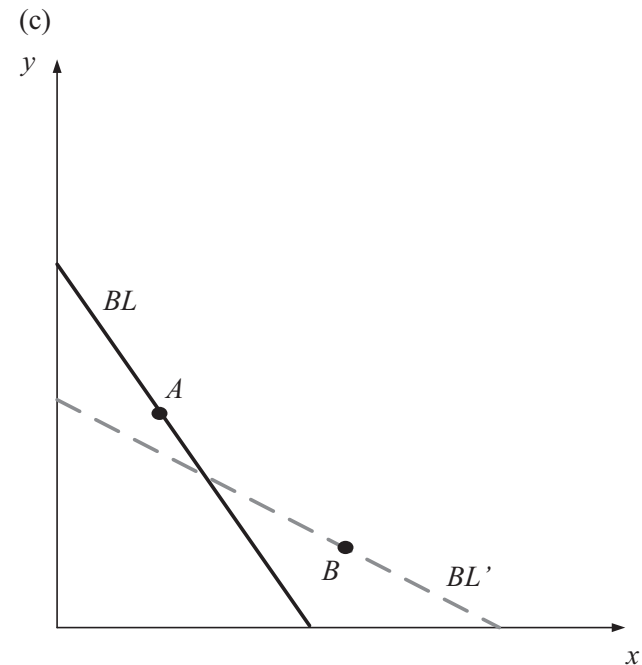
Revealed Preference

- *Example* (continued):
 - Scenario (c). WARP *is not violated*.

Step 1 does not hold.

Bundle A is affordable under BL but B is unaffordable:

- A lies on BL .
- B lies strictly above BL .



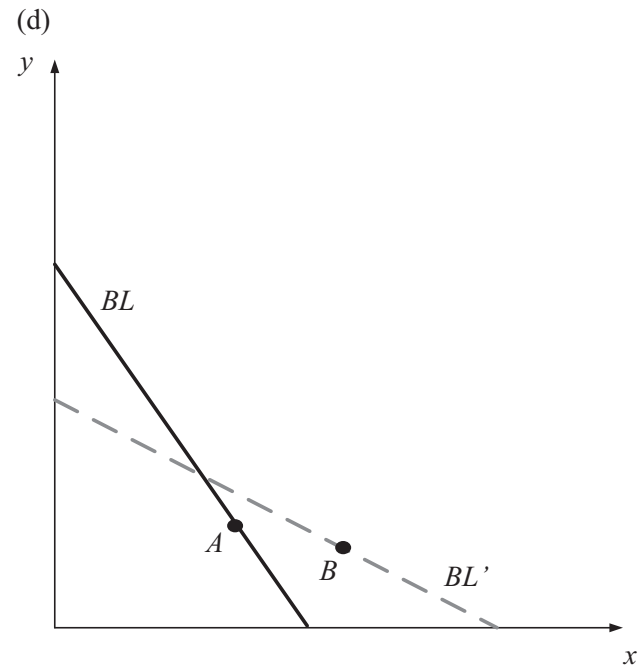
Revealed Preference

- *Example* (continued):
 - Scenario (d). WARP *is not violated*.

Step 1 does not hold.

Bundle A is affordable under BL but B is unaffordable :

- A lies on BL .
- B lies strictly above BL .



Kinked Budget Lines

Quantity Discounts

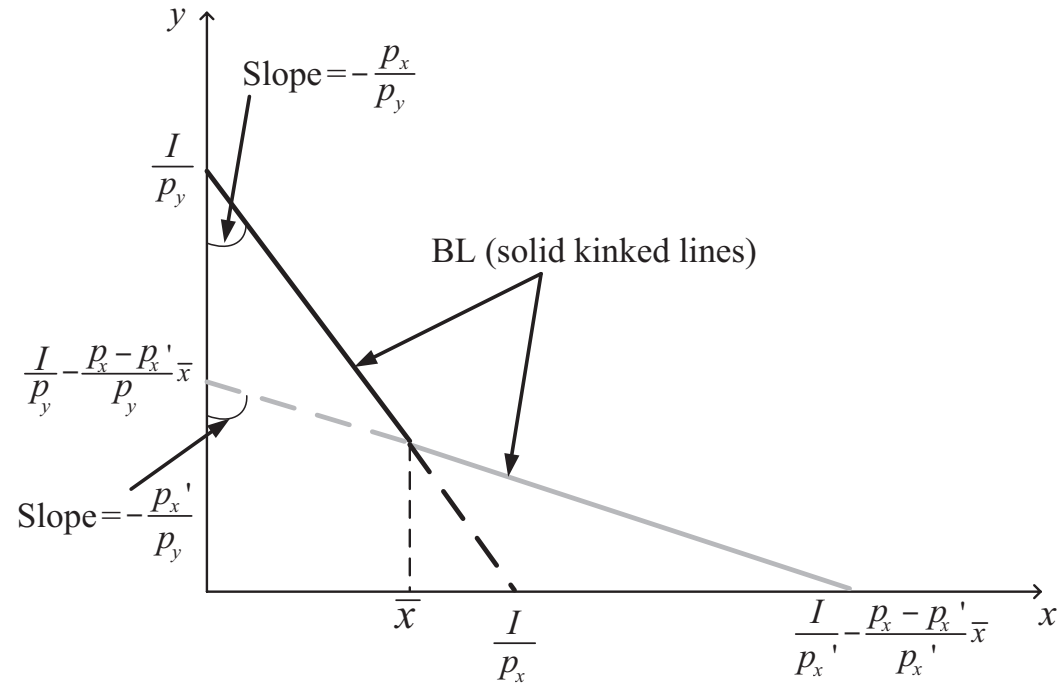
- Sellers offer quantity discounts making first units more expensive than each unit afterwards.

Formally,

- the consumer faces a price p_x for all units of x between 0 and $\bar{x}$ (i.e., for all $x \leq \bar{x}$);
- but she faces a lower price p'_x , where $p'_x < p_x$, for each subsequent unit (i.e., for all $x > \bar{x}$).

Quantity Discounts

- Graphically,



Quantity Discounts

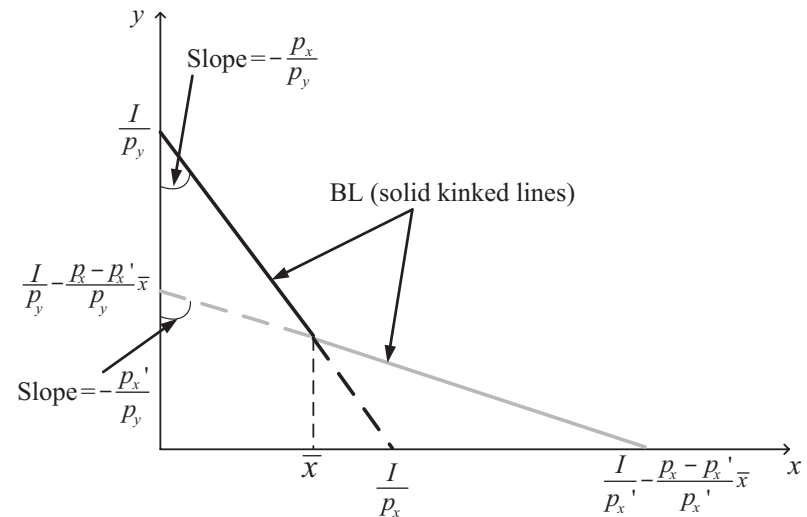
- Mathematically, the equation of the budget line is
 - For all $x \leq \bar{x}$,

$$y = \underbrace{\frac{I}{p_y}}_{\text{Vertical intercept}} - \underbrace{\frac{p_x}{p_y} x}_{\text{Slope}}.$$

- For all $x > \bar{x}$,

$$y = \underbrace{\left(\frac{I}{p_y} - \frac{p_x - p'_x}{p_y} \bar{x} \right)}_{\text{Vertical intercept}} - \underbrace{\frac{p'_x}{p_y} x}_{\text{Slope}}.$$

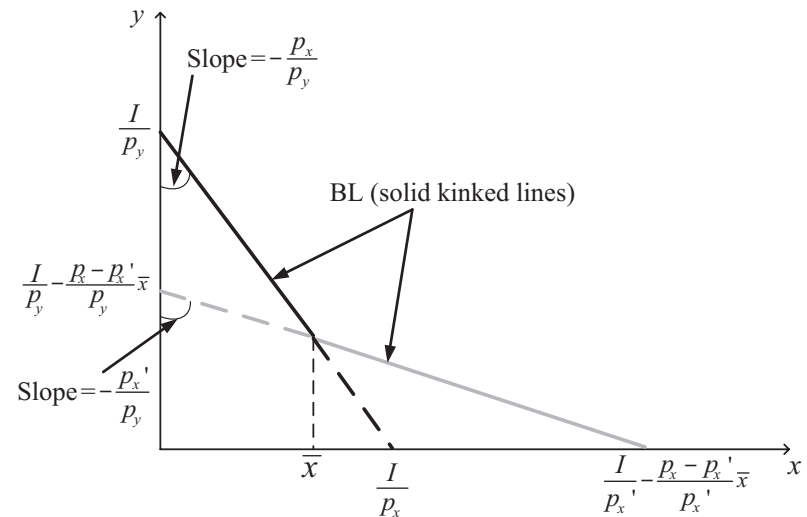
Note $\frac{p'_x}{p_y} < \frac{p_x}{p_y}$ and $\frac{I}{p_y} - \frac{p_x - p'_x}{p_y} \bar{x} < \frac{I}{p_y}$.



Quantity Discounts

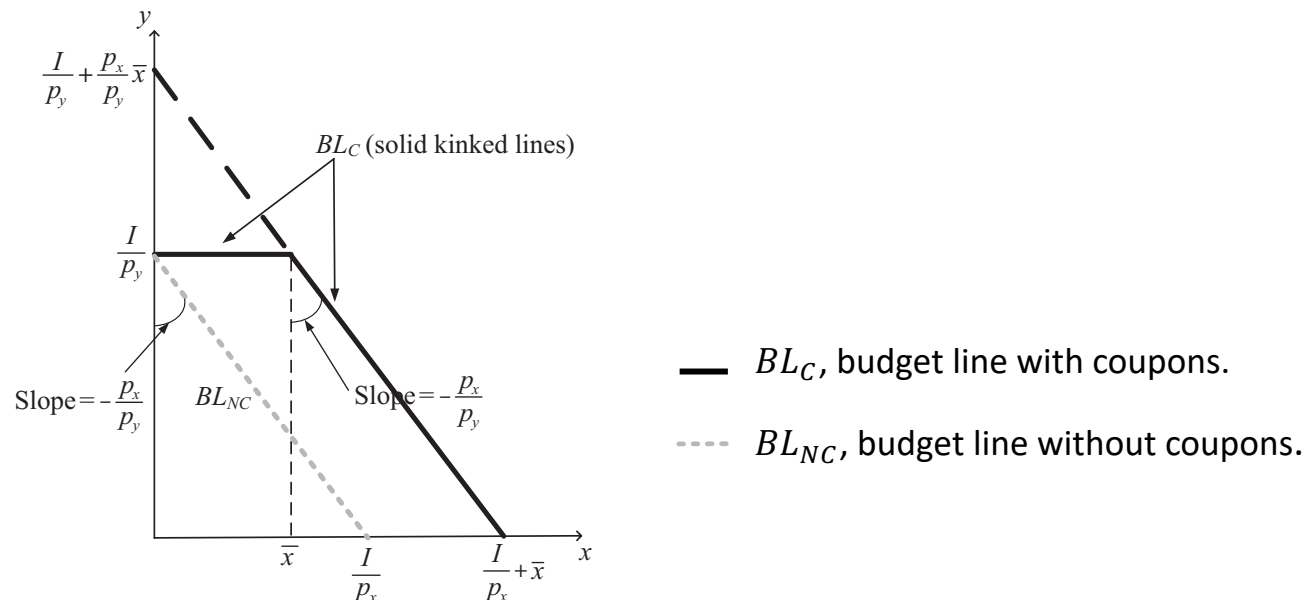
- Effect of a large or small price discount:

- A large discount makes the difference $p_x - p'_x$ larger, shifting the vertical intercept downward and flattening the right segment of the budget line.
- A small discount produces a small difference $p_x - p'_x$, pushing the vertical intercept upward and steepening the right segment of the budget line.



Introducing Coupons

- Consider a market where the government offers coupons, letting consumers purchase the first $\bar{x}$ units of good x for free.



- The coupons expand the set of bundles the consumer can afford.

Introducing Coupons

- Mathematically, this kinked budget line BL_C is

$$BL_C = \begin{cases} p_y y = I \text{ for all } x < \bar{x}, \text{ and} \\ p_x(x - \bar{x}) + p_y y = I \text{ for all } x \geq \bar{x}. \end{cases}$$

Introducing Coupons

- Solving for y , we can represent BL_C as

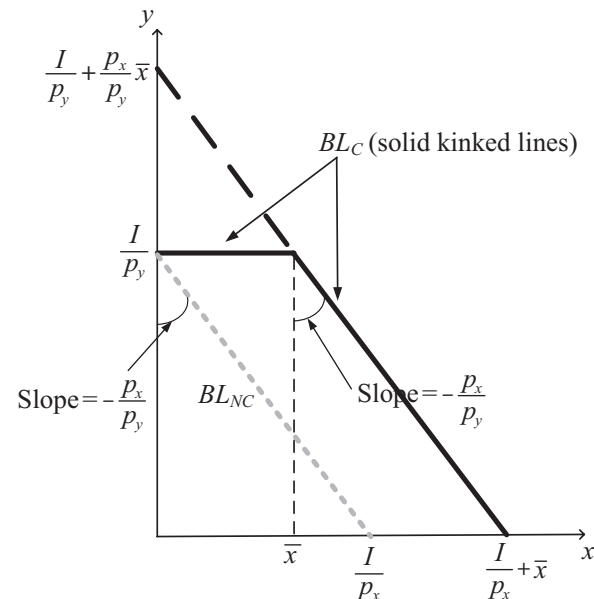
- For $x < \bar{x}$, $y = \frac{I}{p_y}$

- For $x \geq \bar{x}$, $y = \frac{I}{p_y} + \frac{p_x}{p_y} (x - \bar{x})$

or

$$y = \underbrace{\left(\frac{I}{p_y} + \frac{p_x}{p_y} \bar{x} \right)}_{\text{Vertical intercept}} - \underbrace{\frac{p_x}{p_y} x}_{\text{Slope}}$$

Setting $y = 0$, and solving for x , we find the horizontal intercept at $x = \frac{I}{p_x} + \bar{x}$.



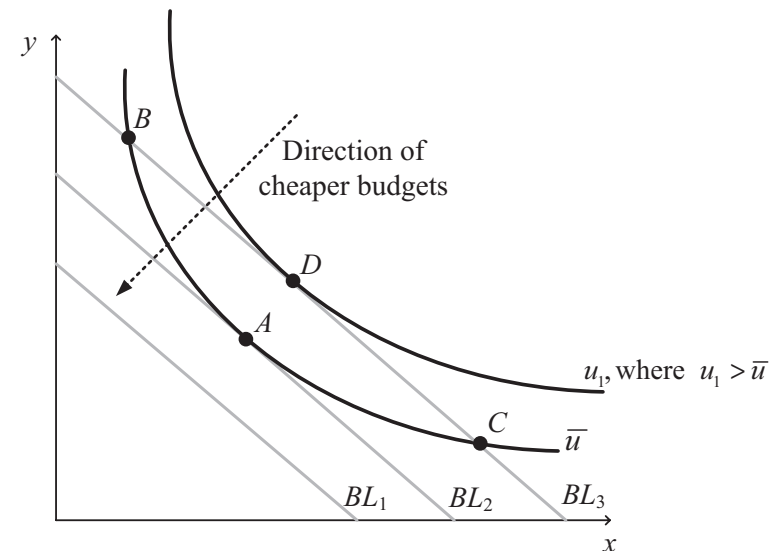
Expenditure Minimization Problem

Expenditure Minimization Problem

- The UMP considers a fixed budget and finds which bundle provides the consumer with the highest utility.
- Alternatively, the consumer could minimize her expenditure while reaching a fixed utility level.
- This is the approach that the **expenditure minimization problem (EMP)** follows.

Expenditure Minimization Problem

- Graphically, the EMP is understood as the consumer seeking to reach an IC with a target utility level $\bar{u}$, but shifting her budget line as close to the origin as possible.
- Bundles B or C cannot be optimal despite reaching $\bar{u}$. She spends more income than in A .
- Bundle D cannot be optimal. She can find cheaper bundles and reach $\bar{u}$.



Expenditure Minimization Problem

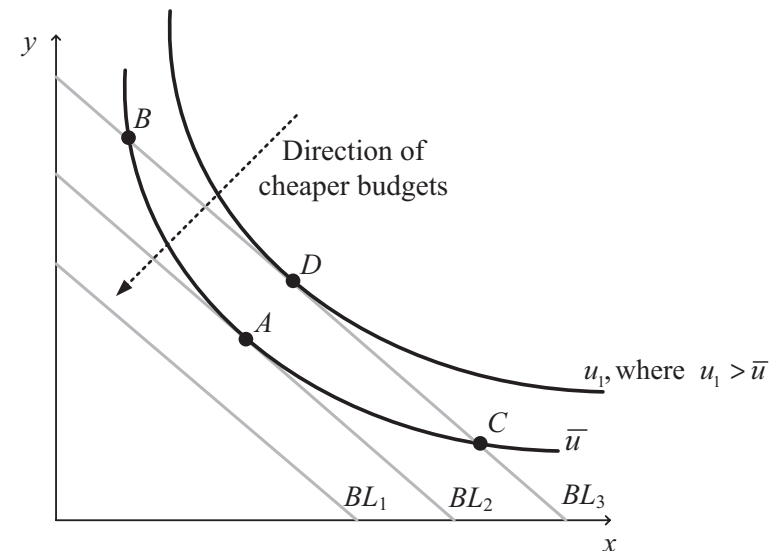
- Bundle A must be optimal. There are no other bundles reaching a lower expenditure than BL_2 .

At A , the indifference curve and the budget line are tangent,

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y}.$$

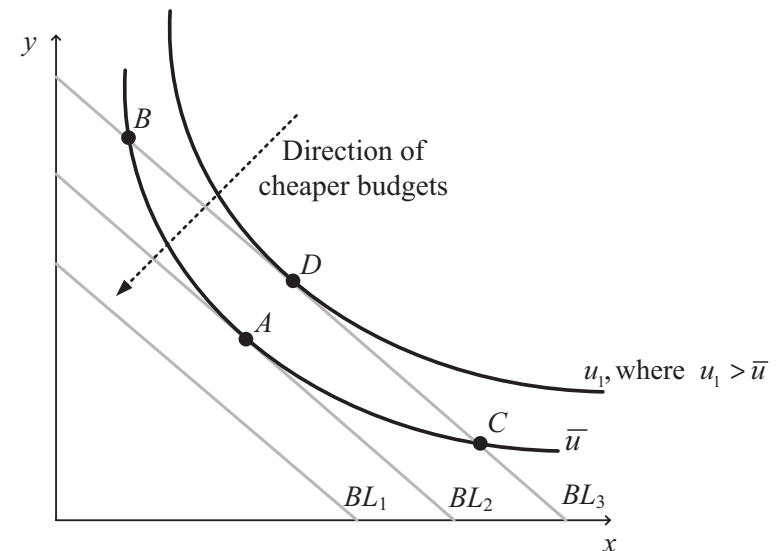
Her constraint is $u(x, y) \geq \bar{u}$, but she would always choose at

$u(x, y) = \bar{u}$. She would never choose bundle satisfying $u(x, y) > \bar{u}$, such as D .



Expenditure Minimization Problem

- Bundle D cannot be optimal. She can find cheaper bundles and reach $\bar{u}$. These bundles that still satisfy the constraint and can be purchased at lower cost.



Expenditure Minimization Problem

- *Procedure to solve the EMP:*

1. Set the tangency condition as $\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$. Cross-multiply and simplify.
2. If the expression for the tangency condition:
 - a. Contains both unknowns (x and y), solve for y , and insert the resulting expression into the utility constraint $u(x, y) = \bar{u}$.
 - b. Contains only one unknown (x or y), solve for that unknown, and insert the result into the utility constraint $u(x, y) = \bar{u}$.

Expenditure Minimization Problem

- *Procedure to solve the EMP* (cont.):

2. If the expression for the tangency condition:

c. Contains no good x or y , compare $\frac{MU_x}{p_x}$ against $\frac{MU_y}{p_y}$.

- If $\frac{MU_x}{p_x} > \frac{MU_y}{p_y}$, set good $y = 0$ in the utility constraint and solve for good x
- If $\frac{MU_x}{p_x} < \frac{MU_y}{p_y}$, set $x = 0$ in the utility constraint and solve for y .

Expenditure Minimization Problem

- *Procedure to solve the EMP* (cont.):
 3. If, in step 2, you find that one of the goods is consumed in negative amounts (e.g., $x = -2$), then set the amount of this good equal to 0 on the utility constraint (e.g., $u(0, y) = \bar{u}$), and solve for the remaining good.
 4. If you haven't found the values for all the unknowns, use the tangency conditions from step 1 to find the remaining unknown.

Expenditure Minimization Problem

- *Example: EMP with a Cobb-Douglas utility function.*
 - Consider an individual with Cobb-Douglas utility function

$$u(x, y) = x^{\frac{1}{3}}y^{\frac{2}{3}},$$

facing $p_x = 10$, $p_y = 20$, and a utility target $\bar{u}$.

- We seek to apply the tangency condition, $\frac{MU_x}{Mu_y} = \frac{p_x}{p_y}$. We first need to find $\frac{MU_x}{Mu_y}$,

$$\frac{MU_x}{Mu_y} = \frac{\frac{1}{3}x^{-\frac{2}{3}}y^{\frac{2}{3}}}{\frac{2}{3}x^{\frac{1}{3}}y^{-\frac{1}{3}}} = \frac{y}{2x}.$$

Expenditure Minimization Problem

- *Example* (continued):

- *Step 1.* The tangency condition reduces to

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y},$$
$$\frac{y}{2x} = \frac{10}{20} \implies y = x.$$

This result contains both x and y , so we move to step 2a.

- *Step 2a.* The utility constraint $u(x, y) = \bar{u}$ becomes $x^{\frac{1}{3}}y^{\frac{2}{3}} = \bar{u}$.
Inserting $y = x$,

$$x^{\frac{1}{3}} \underbrace{(x)^{\frac{2}{3}}}_y = \bar{u} \implies x = \bar{u}.$$

For instance, if $\bar{u} = 5$, the optimal amount of x is $x = 5$.

Expenditure Minimization Problem

- *Example* (continued):

Because we found a positive amount of good x , we move to step 4.

- *Step 4.* Using the tangency condition, $y = x$,

$$y = \bar{u}.$$

- *Summary.* The optimal consumption bundle is $x = y = \bar{u}$, consuming the same amount of each.

Expenditure Minimization Problem

- *Example:*

- Consider the demand from an earlier example

$$u(x, y) = xy + 7x,$$

facing $p_x = 1$, $p_y = 2$, and a utility target $\bar{u} = 70$.

- *Step 1.* The tangency condition reduces is

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y},$$
$$\frac{y + 7}{x} = \frac{1}{2} \implies 2y + 14 = x.$$

This result contains both x and y , so we move to step 2a.

Expenditure Minimization Problem

- *Example* (continued):

- *Step 2a.* Inserting the result from the tangency condition, $2y + 14 = x$, into the utility target $xy + 7x = 70$,

$$\underbrace{(2y + 14)y}_x + 7\underbrace{(2y + 14)}_x = 70,$$

$$2(7 + y)^2 = 70 \Rightarrow (7 + y)^2 = 35,$$

$$\sqrt{(7 + y)^2} = \sqrt{35} \Rightarrow 7 + y = \sqrt{35},$$

$$y = -1.08 \text{ units.}$$

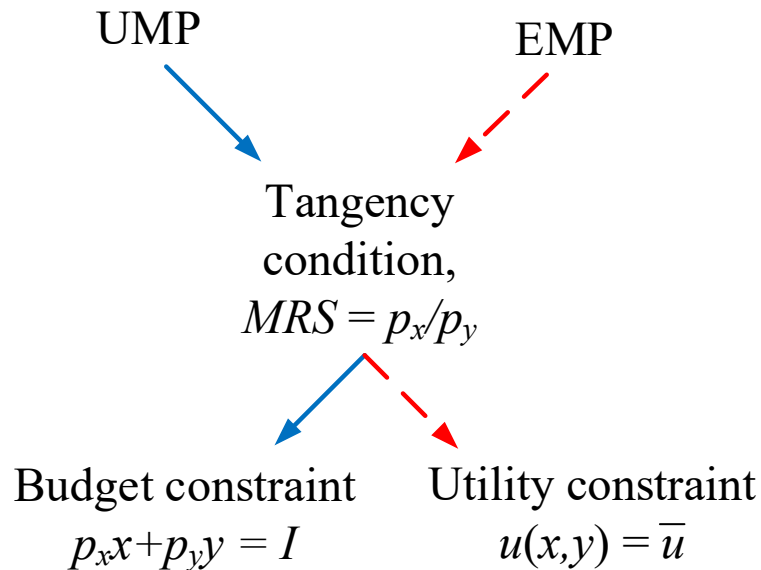
Because we found negative units of at least one good, we need to apply step 3 next.

Expenditure Minimization Problem

- *Example* (continued):
 - *Step 3.* The individual consumes 0 amounts of y , and dedicates all her income to buy x . Because $y = 0$, her utility constraint becomes $u(x, 0) = 70$, or
$$x \cdot 0 + 7x = 70,$$
$$x = 10 \text{ units.}$$
 - *Summary.* The optimal consumption bundle is $x = 10$ and $y = 0$.

Relationship between UMP and EMP

- Similarities and differences of UMP and EMP approaches:



Relationship between UMP and EMP

- Both approaches lead as to the same optimal consumption bundle. The EMP is dual representation of UMP.
- Consider a consumer that solves her UMP and finds optimal bundle

$$(x^U, y^U).$$

- In this situation, the utility she can reach when purchasing this bundle is

$$u(x^U, y^U).$$

Relationship between UMP and EMP

- If we ask the consumer to solve her EMP to reach a target utility level of

$$u(x^U, y^U) = \bar{u},$$

the bundle that solves her EMP coincides with that of UMP.

- We can draw the opposite relationship, starting from EMP.
- Let (x^E, y^E) be the optimal bundle solving EMP.

Relationship between UMP and EMP

- Let I^E be the income the consumer needs to purchase her optimal bundle (i.e., $p_x x^E + p_y y^E = I^E$).
- If we ask her to solve her UMP, giving an income of $I = I^E$, the optimal bundles solving her UMP,

$$(x^U, y^U),$$

coincides with that solving her EMP,

$$(x^E, y^E).$$